THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 6: **Heat Transfer in Nature** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In deserts, nights are cold and days are hot because sand: 1

(a) heats and cools quickly **(b)** heats slowly
(c) stores heat **(d)** reflects all

2. A chimney is made tall so that: 1

(a) hot gases rise and draw fresh air **(b)** bricks cool
(c) rain cannot enter **(d)** smoke is hidden

3. A wall painted white keeps the room cooler than a wall painted black because: 1

(a) white reflects most radiation **(b)** black reflects
(c) white conducts **(d)** white absorbs

4. Water in tanks near the sea stays warm at night because: 1

(a) water is heated by the moon **(b)** water cools slowly
(c) water is hot **(d)** wind is warm

5. A house in a hot area should have windows high up because: 1

(a) dust is kept out **(b)** hot air can escape
(c) light enters **(d)** cold air escapes

6. At the same temperature, a metal chair feels colder than a wooden chair because: 1

(a) metal is colder
(b) wood is hotter
(c) metal conducts heat away from the body faster
(d) wood is a conductor

7. Clouds at night keep the ground warmer because: 1

(a) they cool the ground **(b)** they heat the sun
(c) they carry light **(d)** they reduce loss of radiation

8. The temperature near the ceiling of a closed room with a heater is: 1

(a) lower **(b)** same
(c) higher than near the floor **(d)** zero

9. In winter, an iron railing feels colder than a wooden bench. If both are at 5 °C, a thermometer would show: 1

(a) nothing **(b)** lower for iron
(c) lower for wood **(d)** the same temperature

10. Clear nights are colder than cloudy nights because: 1

(a) heat radiates out unhindered **(b)** air is hot

(c) clouds are cold **(d)** stars are cold

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are cooking vessels sometimes blackened at the bottom? 2

12. Why does a cold glass bottle feel colder than a cold plastic bottle at the same temperature? 2

13. Explain why desert nights are cold though days are hot. 2

14. Why do fire-fighters wear shiny suits? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Design an experiment to prove that heat travels through a vacuum. 3

16. Why do firemen cut a wall of vegetation as a firebreak? 3

17. Explain why electric kettles have heating elements near the bottom. 3

OR

Explain the role of land and sea breezes on the climate of coastal cities. 3

18. Explain why clothes dry faster in sunshine than in the shade. 3

OR

Why does a closed car parked in the sun become hot inside? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Discuss the heat transfer when you boil water in a metal vessel on a gas stove: flame to vessel, vessel to water, water to air. 5

OR

Evaluate whether it is better to use a metal or a ceramic pot for slow cooking and for fast cooking. 5

20. Explain the heating of the Earth by the Sun and the role of the atmosphere, land and sea in producing winds. 5

OR

Explain the working of a double-glazed window in cold climates. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND